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Effects of Nightly Electronic Device Restriction and Morning Bright Light on Sleep, Mood, and Performance Among Elite Athletes

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Abstract

Hoshikawa, M. Effects of nightly electronic device restriction and morning bright light on sleep, mood, and performance among elite athletes. *J Strength Cond Res* 39(11): 1155–1162, 2025—This study aimed to evaluate whether restriction of electronic device use before bedtime and morning bright-light exposure influences sleep, sleepiness, mood, and performance parameters among elite athletes. Twenty-five elite athletes (24 international level and 1 national level) were divided into 2 groups: with (INT; $n = 12$) and without (CON; $n = 13$) intervention. The INT turned off their electronic devices 30 minutes before bedtime and were exposed to morning bright light (approximately 10,000 lux) for 20 minutes after waking. Their nocturnal sleep parameters, Karolinska sleepiness scale, subjective sleep quality, fatigue, and heart rate immediately after waking were monitored every morning for 2 weeks. The Epworth sleepiness scale (ESS) score, Profile of Mood States (POMS) vigor and fatigue scores, reaction time, isometric grip strength, squat jump (SJ) height, rebound jump (RJ) height, RJ index, and heart rate during aerobic exercise were evaluated before and after the 2-week period. The RJ index was calculated as RJ height (m)/ground contact time (sec). For the INT, POMS vigor score ($p < 0.001$), reaction time ($p = 0.016$), RJ height ($p = 0.001$), and RJ index ($p < 0.001$) improved without any change in sleep parameters. The improvements in ESS score and SJ height seemed to be greater in the INT group, but the interactions did not reach a significant level ($p < 0.10$). No statistically significant interaction was observed in the exercise heart rate and isometric grip strength. Two weeks of 30 minutes nighttime electronic device restriction and 20 minutes morning bright-light exposure improved mood and some performance parameters among elite athletes. However, the intervention did not change athletes' sleep parameters.

Key Words: light therapy, intervention, vigor, reaction time, rebound jump height and index

Introduction

To improve athletic performance, competitive athletes train with high frequency, volume, and intensity. For example, 30% of elite Japanese athletes complete >10 training sessions per week (19). Fatigue can compromise training success, and fatigue management is crucial when increasing training load and frequency. Sleep is an important aspect of athletic recovery and regeneration, and nocturnal sleep is known to influence daytime alertness, mood, cognitive function, and sports performance (34,35,43,46). However, recent studies have revealed that many athletes do not sleep enough (29,30). Sargent et al. (45) reported that early morning training severely restricts the amount of sleep for elite swimmers. Many athletes sleep longer and wake later on free days to address loss of sleep on training days. Sargent et al. (45) also reported that swimmers slept for 1.7 hours longer on free days than on training days, with delayed times of going to bed and waking. Hoshikawa et al. (19) also found that half of 817 elite athletes did not have regular wake times (within 2 hours).

Despite the desire for a regular sleep schedule, in reality, athletes' sleep schedules vary depending on their daily training/competition schedules. For instance, early morning or late evening training sessions may be scheduled based on various factors outside the athlete's control (e.g., availability of facilities and training partners, extreme daytime heat, and work/study schedules). Thus, irregular sleep schedules are common even if athletes recognize the need for a regular sleep schedule.

An irregular sleep schedule affects circadian rhythm parameters and melatonin secretion rhythm. Taylor et al. (49) reported that a 3-h delay in waking on a weekend alters the melatonin secretion rhythm and significantly increases daytime sleepiness, which may last until the middle of the following week. In addition, an irregular sleep schedule may affect mood. Recent studies have revealed that depressive mood is associated with the difference between the midpoints of sleep duration on weekdays and weekends (i.e., social jet lag) (31). Therefore, irregular sleep schedules may affect athletes' daytime sleepiness, mood, and performance.

Bright light is the main external influencer of the biologic clock, which is situated in the suprachiasmatic nucleus (12). Based on the phase–response curve describing the relationship between light exposure and circadian rhythm (36), morning bright-light exposure helps advance the circadian system. Burgess et al. (5) reported that 3.5 hours morning bright light and advancing schedule 1-hour a day for 3 days produced a phase advance of 2 hours. Rosenthal et al. (44) reported that 2 weeks of daily morning bright-light exposure (2,500 lux from 7:00 to 9:00) helped to advance sleep schedules. In addition, neural pathways from the suprachiasmatic nucleus to the pineal gland transfer photic information to suppress melatonin secretion (37). Furthermore, bright light has alerting (7,9) and antidepressant (32) effects. Although light therapy is widely accepted as an effective

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practice, there are no standardized guidelines regarding the duration, intensity, and timing of light exposure. The most commonly recommended regimen involves broad-spectrum bright light at 2,000–10,000 lux for approximately 1–3 hours during the early morning (6:00–8:00) (2,11). However, busy athletes may not be able to spare their 1–3 hours every morning to undergo light exposure.

Because light, especially blue light, suppresses melatonin secretion and increases alertness, a number of studies have focused on the effects of blue light exposure on sleep. Silvani et al. (47) summarized 36 studies, reporting that one-third noted decreased sleep duration, more than two-thirds showed increased alertness, and slightly less than half reported increased sleep latency and decreased sleep efficiency with the use of blue light at night. Therefore, morning bright-light exposure and nightly blue-light restrictions may have positive effects on sleep, mood, and performance among elite athletes.

In the review of sleep hygiene for optimizing recovery in athletes, Vitale et al. (50) recommended bright light on awakening and avoiding blue light for at least 2 hours before bed. However, Knufineke et al. (25) combined restricting blue light for 3 hours before bedtime with morning bright light for 1 week and reported shortened subjective, but not objective, sleep latency. Flower et al. (16) examined the effects of combining 1 hour of restriction on electronic device use before bed with 3 hours of exposure to artificial morning blue light (through light goggles). Their intervention lasted from 2 days before to 2 days after a simulated international flight, and the subjects' sleep, mood, and performance were evaluated. They found that the interventions improved feelings of vigor, but had no effect on performance. Thus, despite a growing number of studies, evidence supporting light regulation to optimize sleep and performance in elite athletes remains insufficient.

In this study, as compared with previous studies (2,5,11,16,25,44), I set a shorter intervention duration in a 24-hour period, stronger morning light intensity, and longer intervention period. By doing so, I believe that athletes can easily continue it. This study aimed to examine whether restriction of electronic device use for 30 minutes before bedtime and 20 minutes of morning bright-light exposure for 2 weeks improved athletes' sleep, daytime sleepiness, mood, and performance. If successful, it can be used as a remedy to improve sleep, mood, and/or daytime function in athletes.

Methods

Experimental Approach to the Problem

This study aimed to evaluate whether restriction of electronic device use before bedtime and morning bright-light exposure influences sleep, sleepiness, mood, and performance parameters among elite athletes. The athletes were assigned to either the intervention group (INT) or the control group (CON). For the 2-week experimental period, the INT turned off their electronic devices 30 minutes before bedtime and were exposed to morning bright light (approximately 10,000 lux) for 20 minutes after waking. Sleep parameters during the experimental period as well as mood and performance before and after the period were evaluated.

Subjects

Thirty-eight athletes (34 international and 4 national level, 19 men and women each) were initially recruited at Japan Institute of Sports Sciences for this study. They were randomly assigned to the INT or CON group, with 19 members each. However, owing to pain, injuries, illness, alteration of daily schedule because of close contact with COVID-19–infected persons, and schedule conflicts, 13 athletes were excluded, leaving 25 athletes (24 international and 1 national level, 11 men and 14 women). As a result, 12 athletes (5 men and 7 women) remained in the INT group, and 13 athletes remained in the CON group (6 men and 7 women). They competed in fencing, swimming, taekwondo, diving, gymnastics, wrestling, bicycle racing (track), climbing, canoeing, short-track skating, Nordic combined skiing, and free-style skiing. The athletes' ages and anthropometric data are presented in Table 1. None of the athletes had traveled abroad within the 2 weeks before the commencement of the study, and none were diagnosed with sleep disturbances or using any medications. They continued the usual training during the 2-week experimental period. However, none of the subjects exercised strenuously just before the evaluation of daytime sleepiness, mood, and performance before and after the experimental period.

All subjects provided written informed consent after receiving a full explanation of the experimental procedure and the possible risks. Written informed consent was also obtained from the parents of athletes who were <20 years of age (2 men and 3 women), and they received a full explanation of the experimental procedure and possible risks. The study protocol was approved by the Japan Institute of Sports Sciences ethics committee (approval number: 017).

Questionnaires to Evaluate the Athletes' Characteristics Before the Experiment

Before the experiment, to clarify their characteristics, the athletes were asked to complete questionnaires regarding their training schedules, the Pittsburgh Sleep Quality Index (PSQI) (6), and a reduced version of the Morningness-Eveningness Questionnaire (rMEQ) (18). The PSQI consists of 19 self-administered questions, and subjective sleep quality was evaluated using the global PSQI score; a global score of >5.5 in the Japanese version indicates poor sleep quality (13). The rMEQ, developed by Adam and Almirall (1), consists of 5 items extracted from the original 19 items in the Morningness-Eveningness Questionnaire (18). The validity and reproducibility of the Japanese version have been previously verified (39). The rMEQ scores range from 4 to 25, and the scores are classified as 4–7, "definitely evening type"; 8–11, "moderately evening type"; 12–17, "neither type"; 18–21, "moderately morning type"; and 22–25 "definitely morning type." In addition to the PSQI and rMEQ, the athletes were asked about their bedtime and wake-up time on both training days and free days.

Study Design

In both groups, nocturnal sleep was monitored using an actigraph (Actiwatch Spectrum; Respironics, Inc.) and self-reported data

Table 1
Athletes' age and anthropometric data.*

	Men (n = 11) Mean ± *SD*	Women (n = 14) Mean ± *SD*
Age (yrs)	26.3 ± 6.3	21.9 ± 5.1
Height (cm)	178.4 ± 5.6	162.0 ± 5.7
Body mass (kg)	76.6 ± 9.6	56.9 ± 3.5
BMI (kg·m^{-2})	24.0 ± 2.2	21.7 ± 1.3

*Values are reported as mean ± *SD*. BMI = body mass index.

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regarding subjective sleep quality, sleepiness, fatigue, and heart rate immediately after waking for a 2-week experimental period. Before and after the 2 weeks, daytime sleepiness, mood, and performance were evaluated.

Intervention

For the INT group, the athletes were instructed to turn off electronic devices, including their smartphones, at individually set times. The time was set at 30 minutes before bedtime on the training day when they completed the questionnaire. In addition, they were exposed to bright light for 20 minutes immediately after waking up. The light (color rendering index: Ra 93, full spectrum) was produced using a bright light ME+Pro device (Over the Top Ltd., Japan; 53.5 cm tall × 44 cm wide), which was placed 35 cm away from the athlete. The device can provide bright light of approximately 10,000 lux when the distance is set to 35 cm. The CON group did not undergo any of these interventions. In the INT group, the athletes had to notify the author through the electric form in the morning if they had not been exposed to morning bright light or had used electronic devices after the individual set time on the previous day.

Daily Monitoring of Heart Rate, Sleepiness, Subjective Sleep Quality, Fatigue, and Nap Duration

During the 2 weeks, all athletes measured their heart rate (Pulsox-1, Konika Minolta, Tokyo, Japan) immediately after waking and while still in bed. Next, they completed an electronic form that collected data regarding their bedtime and wake-up time, and self-reported values for sleepiness, sleep quality, fatigue, heart rate, and nap duration. The athletes then sent the form to the author, at which point they could not review or modify their responses. Subjective sleepiness was evaluated using the Karolinska sleepiness scale (KSS), Japanese version (23). Subjective sleep quality was assessed using a 10-point Likert scale, with a score of 1 indicating extremely poor sleep quality and a score of 10 indicating very good sleep quality. Self-reported fatigue was assessed using the scale developed and recommended by the Japanese Society of Fatigue Science, with a score of 1 indicating "completely refreshed" and 10 indicating "completely fatigued" (21).

Daily Sleep Monitoring

During the experimental period for both groups, each athlete's sleep was monitored using a wristwatch-based actigraph device with a marker button and a light sensor. The athletes pressed the marker button when they were going to sleep and when they woke. Every night, the time at which the marker button was pressed and the time when their room lights were switched off were compared; bedtime was determined as whichever was later between these 2 timings. The actigraphy data during 1-minute epochs were scored as sleep or wake using the related software (Actiware Ver. 6.0.1; Respironics, Inc.). The medium threshold was set at 40 activity counts.

The sleep data of each night were analyzed to determine the time in bed, sleep latency, wake after sleep onset, total sleep time, and sleep efficiency. Time in bed was defined as the period between going to bed and waking. Sleep latency was defined as the amount of time between going to bed and sleep onset. Wake after sleep onset was defined as the amount of time spent awake after sleep onset. Total sleep time was calculated as (time in bed) − (sleep latency + wake after sleep onset). Sleep efficiency was calculated by dividing the total sleep time by the time in bed. Sleep latency and sleep efficiency parameters are the indices of sleep quality.

Daytime Sleepiness, Mood, and Performance Before and After the Experimental Period

Evaluations of daytime sleepiness, mood, and performance were conducted during the daytime (11:00–14:30) immediately before the first night and immediately after the 2-week intervention (in the INT) or monitoring (in the CON) periods. Because our athletes were from various sports, their training schedules varied. Thus, the times for evaluating daytime sleepiness, mood, and performance were set individually between 11:00 and 14:30.

The athletes came to our laboratory at the same hour before and after the 2-week period. They completed the questionnaires on daytime sleepiness and mood states. Subsequently, they participated in the performance tests.

Daytime sleepiness was evaluated using the Epworth sleepiness scale (ESS) (48). An ESS score of >10.5 indicates excessive daytime sleepiness (48). Mood was evaluated using vigor and fatigue scores from the Profile of Mood States (POMS) (53).

The performance tests consisted of a psychomotor vigilance task (PVT), isometric handgrip strength, jump height tests, and heart-rate monitoring during aerobic exercise. The mean reaction time during PVT was evaluated using a handheld PVT monitor (PVT-192; AMI, New York). Each athlete watched the LED timer on the monitor and pressed the response button with their right thumb as quickly as possible after the appearance of the visual stimulus, which appeared at random intervals of 2–10 seconds for 10 minutes. The mean reaction time was analyzed.

Maximum isometric grip strength was measured using an isometric dynamometer (Type YX, Tsutsumi Production Inc., Fukushima, Japan). In Japan, elementary, junior high, and high schools under the jurisdiction of the Ministry of Education, Culture, Sports, Science, and Technology conduct physical fitness tests annually for all students, except for those who have been restricted from exercise by a doctor and those who complain of unwell (fever, pain, etc.). Isometric grip strength is one of the items measured. Therefore, the measurement protocol was standardized by the ministry, and all athletes were familiar with it. The width of the dynamometer grip was adjusted such that the second joint of the athlete's index finger was approximately 90°. The athlete stood shoulder-width apart, holding the dynamometer in his/her hand so that its gauge needle was on the outside, arms were naturally lowered, and the dynamometer did not touch their body or clothing. The athlete squeezed the dynamometer grip with maximum effort. During the measurement, the athlete was requested to maintain a constant hand position. The measurements were performed twice, alternating between left and right, with the largest value obtained from the 2 repetitions selected for the analysis.

Height values for the squat jump (SJ) and rebound jump (RJ) were evaluated using a switch mat (66 × 100 cm, Multi-JumpTesterII, DKH, Tokyo, Japan), which contains embedded switches that measure flight and ground contact time. Data were sampled at a frequency of 1,000 Hz, and the jump height was calculated using the flight time. For both jumps, the athlete stood

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on the switch mat while keeping his/her hands on his/her hips. The starting position for the SJs was standardized at a 90-degree knee angle. More than 1 second after maintaining a quiet starting position, the athlete jumped as high as possible without any countermovement. The SJ height was measured twice, and the higher values were selected for analysis. For RJ, the athlete performed 6 consecutive vertical jumps (5 RJs) as high as possible while minimizing ground contact time. Each athlete performed 2 repetitions of 5 successive RJs. Rebound jump index (RJ-index), a performance indicator for lower limb stretch shortening cycle ability, was calculated as RJ height (m)/ground contact time (sec). The highest RJ-index and RJ height at that time were used for the analysis. These measurements, based on protocols published by the Japan Institute of Sports Sciences, are commonly used to assess the physical performance of elite Japanese athletes (20).

Heart rate was measured during 5 minutes of aerobic exercise on an electrically braked ergometer (Powermax VII, Combi Co., Ltd., Tokyo, Japan), with intensities of 2.5 kp for men and 1.5 kp for women at a rotation rate of 70 rpm. Heart rate was measured using a heart rate monitor (Polar M200, Polar Electro Oy, Kempele, Finland), and the value from the final 10 seconds was selected for the analysis.

Statistical Analysis

Data are expressed as mean ± *SD*. Data normality was assessed using the Shapiro–Wilk test. Except for repeated measurement data, the unpaired or paired *t* test, Wilcoxon signed-rank test, and Mann–Whitney *U* test were used depending on data normality and whether the data were paired or not. For repeated measurement data, non-normally distributed data were transformed to normally distributed data using Box-Cox transformation. Two-way repeated measures analysis of variance (ANOVA) was used to analyze the effects of day, group, and day × group interactions. A post hoc analysis was performed using Bonferroni's method. All analyses were performed using the BellCurve for Excel statistical package (Social Survey Research Information Co., Ltd., Tokyo, Japan). Differences were considered statistically significant at *p*-values of <0.05.

Results

The results of the questionnaires on athletes' characteristics before the experiment are summarized in Table 2. The PSQI score, rMEQ score, training schedules, training hours, and sleep schedules on both training day and free day did not differ between groups. Ten of the 25 athletes had a PSQI global score of >5.5 before the experiment. The rMEQ score showed 1 athlete as "moderately evening type," 15 athletes as "neither type," 8 athletes as "moderately morning type," and 1 athlete as "definitely morning type." For both groups, bedtimes and wake-up times were significantly late on free days than on training days.

Figure 1 presents nocturnal sleep parameters, KSS score, subjective sleep quality, fatigue, morning heart rate on waking, and nap duration. Owing to mechanical problems, actigraph data could not be obtained from 1 subject in the INT. Therefore, the data measured using the actigraph from the 11 athletes in the INT are shown in Figure 1. No changes with days were observed in bedtime, wake-up time, time in bed, sleep latency, wake after sleep onset, total sleep time, sleep efficiency, subjective sleep quality, morning heart rate, or nap duration. For the KSS score, statistical analysis revealed an insignificant interaction ($p = 0.30$) and a significant effect of day ($p < 0.001$), with significant changes in both groups ($p = 0.044$ for the INT and $p = 0.010$ for the CON, respectively). For fatigue, the interaction did not reach a significant level ($p = 0.088$), but the effect of day ($p = 0.003$) was significant, especially in the INT ($p < 0.001$), but not in the CON ($p = 0.85$).

Table 3 describes the ESS score, POMS fatigue, and vigor scores, and performance test results before and after the 2-week experimental period. Significant interactions were observed in POMS vigor ($p = 0.042$), reaction time ($p = 0.032$), RJ height ($p = 0.031$), and RJ index ($p = 0.017$). Post hoc tests revealed significant improvements in the INT group in POMS vigor ($p < 0.001$), reaction time ($p = 0.001$), RJ height, and RJ index ($p < 0.001$) after 2 weeks of intervention. The decrease in ESS seemed larger for the INT than for the CON; however, the interaction was not significant ($p = 0.095$). In contrast, the interaction did not reach a significant level for ESS ($p = 0.095$) or SJ ($p = 0.085$),

Table 2
Results of the questionnaires for athlete characteristics before the experiment.*†

	INT (*n* = 12) Mean ± *SD*	Significance within a group (*p*)	CON (*n* = 13) Mean ± *SD*	Significance within a group (*p*)	Significance between groups (*p*)
Pittsburgh sleep quality index score	5.5 ± 1.68		4.6 ± 1.76		0.19
Morning-evening questionnaire (reduced scale) score	15.3 ± 2.61		16.5 ± 4.12		0.46
The earliest training-practice start time (h:min)	9:08 ± 1:05		9:19 ± 0:33		0.84
The latest training-practice end time (h:min)	18:40 ± 1:47		17:47 ± 1:23		0.09
Hours of training-practice per week (h:min)	23:30 ± 7:49		21:32 ± 7:11		0.81
Bedtime on training days	23:17 ± 0:56	0.039‡	23:27 ± 0:44	0.007§	0.71
Bedtime on free days	23:40 ± 0:45		23:58 ± 0:53		0.36
Difference in bedtime between training days and free days	0:23 ± 0:36		0:31 ± 0:26		0.23
Wake-up time on training days	7:02 ± 0:47	0.005§	7:19 ± 0:51	0.002§	0.47
Wake-up time on free days	8:32 ± 1:25		8:25 ± 1:28		0.85
Difference in wake-up time between training days and free days	1:30 ± 1:14		1:07 ± 0:56		0.53

*Values are reported as mean ± *SD*.
†CON = control group; INT = intervention group.
‡$p < 0.05$ within the group.
§$p < 0.01$ within the group.

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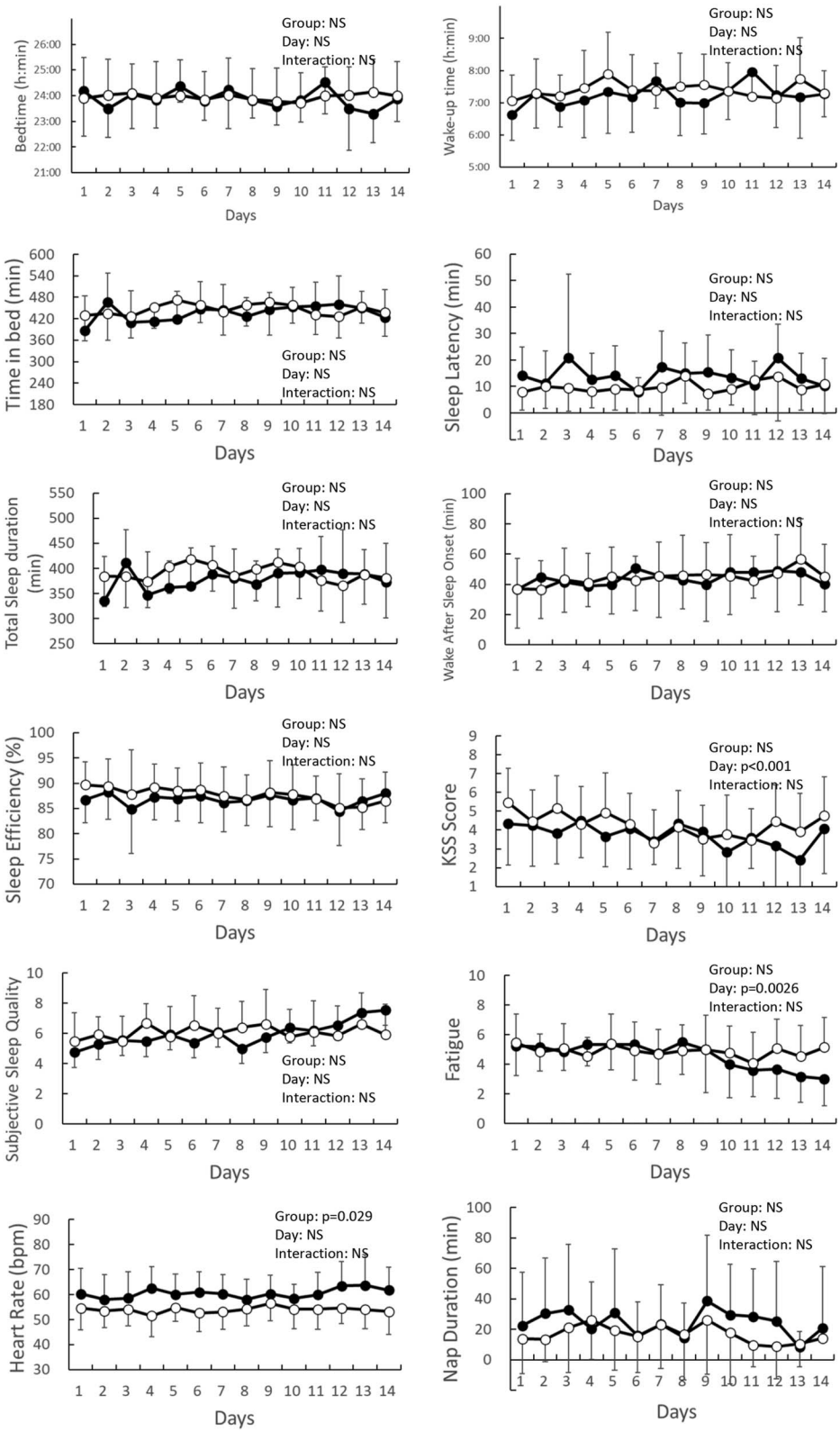

Figure 1. Nocturnal sleep parameters, Karolinska Sleepiness Scale Score, subjective sleep quality, fatigue, morning heart rate immediately after waking, and nap duration during experimental period. ● intervention group, ○ control group. Values are reported as the mean ± *SD*.

although the changes were larger in the INT than in the CON. No interaction was also observed in grip strength and heart rate during exercise.

Table 4 describes the earliest and latest wake-up and bedtimes during the experimental period for both the groups. These parameters were measured using a wristwatch-based actigraph

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Table 3

Epworth sleepiness scale score, POMS fatigue and vigor scores, and performance test results before and after the 2-week experimental period.†

	INT		Significance within the group (*p*)	CON		Significance within the group (*p*)	Group × day (*p*)
	Pre	Post		Pre	Post		
ESS score	10.0 ± 4.1	5.4 ± 2.4	<0.001‡	8.6 ± 2.4	6.7 ± 3.0	0.019	0.095
POMS fatigue	8.3 ± 4.2	5.5 ± 3.5	0.002§	6.8 ± 3.8	4.5 ± 1.6	0.006	0.82
POMS vigor	14.6 ± 6.6	25.9 ± 11.2	<0.001‡	20.0 ± 7.0	22.3 ± 7.0	0.28	0.042‖
Reaction time, ms	240 ± 14.4	225 ± 20.0**	0.001§	238 ± 17.3	241 ± 18.9	0.42	0.032‖
Right grip strength, kg	42.5 ± 14.8	42.8 ± 14.4	0.70	42.0 ± 9.1	42.4 ± 9.5	0.56	0.92
Left grip strength, kg	39.9 ± 12.1	40.9 ± 12.5	0.037¶	38.0 ± 9.3	39.2 ± 10.3	0.010¶	0.78
Heart rate during exercise, bpm	128 ± 9.6	126 ± 11.8	0.075	122 ± 12.0	123 ± 15.0	0.32	0.16
Squat jump height, cm	30.7 ± 7.1	33.2 ± 6.5	0.016¶	33.0 ± 4.5	33.0 ± 4.0	0.98	0.085
Rebound jump height, cm	28.6 ± 5.2**	30.6 ± 5.0	<0.001‡	30.6 ± 2.3	30.4 ± 2.9	0.72	0.031‖
Rebound jump index	1.56 ± 0.37*	1.73 ± 0.35	<0.001‡	1.80 ± 0.31	1.77 ± 0.23	0.44	0.017‖
Contact time in rebound jump, msec	188 ± 40.7	180 ± 31.3	0.078	178 ± 21.1	174 ± 25.6	0.43	0.61

Values are reported as mean ± *SD*.
*$p < 0.05$ between the groups.
**$p < 0.01$ between the groups.
†CON = control group; INT = intervention group; ESS = Epworth sleepiness scale; POMS = profile of mood states; Pre = preintervention; Post = postintervention.
‡$p < 0.001$ within the group.
§$p < 0.01$ within the group.
‖$p < 0.05$ for the interaction.
¶$p < 0.05$ within the group.

device. Compared with the self-reported sleep schedule (Table 2), the actual irregularity of the schedules seems larger. Although the self-reported sleep schedule did not differ between the groups (Table 2), the earliest wake-up time during the experimental period was earlier in the INT group ($p = 0.012$, Table 4). Thus, the difference between the earliest and the latest wake-up times was greater in the INT group ($p = 0.007$, Table 4).

Discussion

The study findings revealed that combined nightly electronic device restriction and morning bright-light exposure helped to improve mood and some performance parameters, such as reaction time and RJ heights, in elite athletes (Table 3). Many studies have reported that sleep extension improves performance and mood (34,43,46). However, bedtime, wake-up time, time in bed, and sleep duration did not change during the 2 weeks of this study (Figure 1), so the improvements in the INT group may not derive from sleep extension. The athletes' characteristics (PSQI global score, rMEQ score, and training schedules) before the study did not differ between the groups (Tables 2); therefore, it seems that they did not affect the improvements.

As previously mentioned, bright light is not only the main external influencer of the biologic clock (12) but also suppresses melatonin secretion (37) and increases alertness (7,9). The most commonly recommended light therapy regimen involves broad-spectrum bright light at 2,000–10,000 lux for approximately 1–3 hours in the early morning (6:00–8:00) (2,11). In this study, although the exposure duration was 20 minutes, the INT used a device that provided bright light with an upper limit of the range (approximately 10,000 lux) if the distance between the device and athlete was 35 cm. Busy athletes may not have 1–3 hours every morning to undergo light exposure; hence, 20 minutes of exposure is easier to maintain. In this study, none of sleep parameters, including bedtime and sleep latency, changed during the 2-week experimental period (Figure 1). Therefore, 20-minute morning bright light exposure may be too short to objectively demonstrate advances in circadian rhythmicity. This study adopted indices that showed diurnal fluctuations, such as reaction time (24,34), grip strength (42,52), and heart rate (42). Hence, the athletes

Table 4

The earliest and latest wake-up and bedtime during experimental period.*†

	INT (*n* = 11) Mean ± *SD*	CON (*n* = 13) Mean ± *SD*	Significance between the groups (*p*)
Earliest wake-up time	5:35 ± 0:51	6:23 ± 0:32	0.012‡
Latest wake-up time	9:06 ± 0:53	8:38 ± 1:04	0.31
Difference between the earliest and the latest wake-up time	3:30 ± 0:52	2:14 ± 0:58	0.007§
Earliest bedtime	22:31 ± 0:53	22:48 ± 0:45	0.66
Latest bedtime	1:57 ± 1:23	1:13 ± 1:24	0.50
Difference between the earliest and the latest bedtime	3:25 ± 1:41	2:25 ± 1:04	0.19

*Values are reported as mean ± *SD*.
†CON = control group; INT = intervention group.
‡$p < 0.05$ between the groups.
§$p < 0.01$ between the groups.

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came to our laboratory at the same hour before and after the 2-week period and then performed the tests. I believe that circadian rhythmicity did not contribute to the improvement of mood and some performance in the INT.

Light from electronic devices is known to reduce melatonin levels, increase alertness at night, delay the circadian clock, and decrease alertness the following morning (10). Many studies have demonstrated that nightly electrical device usage shortened time in bed and sleep duration (10,47,51). However, in athletes, some studies have reported that electronic device restrictions do not affect sleep duration or efficiency (14,22). In addition, previous studies have reported that melatonin secretion was significantly suppressed when electronic devices were used for 4–5 hours (8,10), but not for 2 hours (15,41). As such, it is possible that electronic device restrictions for 30 minutes before bedtime might be too short to have an impact on subsequent sleep.

For POMS vigor, as described in the introduction, morning bright light has alerting (7,9) and antidepressant (32) effects. Although the most commonly recommended regimen involves approximately 1–3 hours (2,11), Partonen (40) reported that not only 1-hour but also 15 minutes daily exposure to morning bright light (3,300 lux) improved mood and subjective sleepiness in winter depression patients. Although the athletes in this study were not patients, 20 minutes morning bright light exposure might have had similar effects on mood and alertness. It is well known that PVT reaction time reflects alertness (3). Thus, the improvements in mood and some performance observed in the INT group (Table 3) may be because of increased alertness by morning bright light rather than a shift in circadian rhythmicity and/or changes in nocturnal sleep.

Jumping requires both muscle strength and explosive power, the latter of which may be greatly influenced by neuromuscular coordination. The RJ contains 3 phases: preactivation before contact with the ground, eccentric movements during sinking on the ground, and subsequently switching to concentric movements. Because the contact time in RJ did not change in either group (Table 3), the improvements in RJ height and RJ index in the INT group were derived from greater power during unchanged contact time. The eccentric phase for leg extensors may contribute to greater heights, which are associated with increased prior activation (27,28), stored elastic energy (4,27), and contribution of the stretching reflex (26). Such a stretch shortening cycle is observed and used in many sporting movements, such as running, sprinting, and hopping (38).

In contrast, for SJ height, the interaction did not reach a significant level, although the change was larger in the INT than in the CON (Table 3). Although SJ involves dominantly concentric movements, SJ and RJ require muscle strength and explosive power, which may be influenced by neuromuscular coordination. One possible explanation for the interaction not reaching a significant level may be the larger variance in SJ height compared with RJ (Table 3). Another possible explanation may be the instructions for RJs. In this study, the athletes were instructed to perform RJs as high as possible while minimizing ground contact time. This means that the 2 types of jumps (SJ and RJ) focused on different elements.

In contrast, isometric grip strength may be less affected by neuromuscular coordination, although it still reflects the muscle strength. Heart rate during aerobic exercise reflects cardiovascular fitness and/or autonomic nervous system activity. Therefore, the present results suggest that the intervention most likely exerts its effects on performance through neuromuscular function. Recently, Gooderick et al. (17) reported that sleep hygiene interventions improved countermovement jump height but not isometric mid-thigh pull strength. This previous result is partially in line with my results. Heart rate variability is frequently used as a measure of autonomic activity. To accurately measure HRV, the measurement must be performed in the same posture for at least 5 minutes, with a controlled respiration rate and elimination of body movement (33). Measurement of HRV immediately after waking up was not used in this study because it is labor intensive for athletes and is performed outside the author's supervision.

As described in Table 2, the athletes reported shifted sleep schedules between training days and free days at baseline. The actigraph data confirmed these results (Table 4), while the measured irregularities in bedtime and wake time were larger than the self-reported irregularities in the questionnaire. These results suggest that athletes underestimated the irregularity of their sleep schedule.

This study had several limitations. First, the results of this study may reflect athletes' baseline characteristics because they did not have regular sleep schedules (Tables 2 and 4). It is unclear whether the same results could be obtained if athletes without such characteristics participated. However, as described in the Introduction section, irregular sleep schedules are frequently observed among athletes. Second, the athletes' training and practice could not be controlled during the 2-week intervention, while strenuous exercises were excluded before the performance tests. However, this study included athletes from various sports. They completed variable training and practice sessions during the 2 weeks. Therefore, the improved performance after the intervention was not derived from any specific training program. Third, the sample size of this study was relatively small. As described in the Methods section, 38 athletes were initially recruited; however, after excluding athletes with conditions that might have affected the measured values, only 25 athletes remained in this study. Post hoc statistical power analysis performed using G*Power version 3.1.9.2 (Heinrich-Heine-Universität Düsseldorf, Düsseldorf, Germany) revealed that power (1-β error probability) was 0.68 (ANOVA: repeated measures, within-between interaction, 2 groups ×2 measurements, effect size $f = 0.253$, α error probability = 0.05). Therefore, attention should be paid to the possibility of Type II errors. Despite these limitations, this study is the first to demonstrate that short periods of light regulation improve mood and some performance in elite athletes without sleep extension.

Practical Applications

The 2-week intervention, which combined nightly electronic device restriction and morning bright-light exposure, helped to improve mood and some performance parameters. The intervention did not change athletes' sleep parameters. The athletes exhibited irregular sleep, and actual sleep irregularities were larger than the self-reported irregularities. To assess an athlete's sleep habits, it is preferable to evaluate them using an appropriate device rather than self-report.

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